SL Math School: Random Growth Models, etc.
Dyson Brownian Motion

1 Core Problems

Problem 1 (Karlin-McGregor Formula)**.** Let $B = (B_1, \ldots, B_n)$ be n independent Brownian motions on $[0, 1]$ with $B_i(0) = x_i$, where $x_1 < \cdots < x_n$. Let

$$\tau = \inf\{t \geq 0 : B_i(t) = B_j(t) \text{ for some } i \neq j\}.$$

The Karlin-McGregor formula says that, for $y_1 < \cdots < y_n$,

$$\mathbb{P}_x \left(B(t) \in d\overline{y},\ \tau > t\right) = \det\left(p_t(y_j - x_i)\right)_{i,j=1}^{n} d\overline{y},$$

where

$$p_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t}.$$

Prove this formula in the case $n = 2$ by considering an appropriate stopping time and applying the strong Markov property.

(Optional Challenge) Sketch how the argument extends to general n.

Problem 2 (Dyson Brownian Motion Transition Probabilities)**.** Define the Weyl chamber by

$$\mathbb{W}_n = \{\mathbf{x} \in \mathbb{R}^n : x_1 > x_2 > \cdots > x_n\}.$$

Let $B = (B_1, \ldots, B_n) \sim \mathbb{P}_{\mathbf{x}}$ be n independent Brownian motions started from $\mathbf{x} \in \mathbb{W}_n$, and define

$$\tau = \inf\{t \geq 0 : B(t) \notin \mathbb{W}_n\}.$$

You may assume the following survival asymptotic:

$$\begin{aligned} \mathbb{P}_{\mathbf{x}}(\tau > T) &= \int_{\mathbb{W}_n} \det\left(p_T(y_j - x_i)\right)_{i,j=1}^{n} \, d\mathbf{y} \\ &\sim C_n \Delta(\mathbf{x}) T^{-n(n-1)/4} \end{aligned}$$

as $T \to \infty$, where

$$\Delta(\mathbf{x}) = \prod_{i<j}(x_i - x_j), \qquad p_t(r) = \frac{1}{\sqrt{2\pi t}} e^{-r^2/(2t)}.$$

For fixed $R > 0$, let $\mathbb{P}_{\mathbf{x}}^T$ denote the law of B on $[0, R]$ conditioned on $\{\tau > T\}$. Assume that as $T \to \infty$, these laws converge on every compact time interval to a limiting law $\mathbb{P}_{\mathbf{x}}^{\infty}$.

Let $W = (W_1, \ldots, W_n) \sim \mathbb{P}_{\mathbf{x}}^{\infty}$. Show that W has transition density

$$\mathbb{P}_{\mathbf{x}}^{\infty}\left(W_{t+s} \in d\mathbf{z} \,|\, W_t = \mathbf{u}\right) = \det\left(p_s(z_j - u_i)\right)_{i,j=1}^{n} \frac{\Delta(\mathbf{z})}{\Delta(\mathbf{u})}$$

for $\mathbf{u}, \mathbf{z} \in \mathbb{W}_n$ and $s > 0$.

Remark: We refer to $\mathbb{P}_{\mathbf{x}}^{\infty}$ as the law of Dyson Brownian motion started from $\mathbf{x}$.

2 Extra Problems

Problem 3 (Lindström-Gessel-Viennot)**.** Let G be a finite directed graph with no cycles and edge weights $w(e)$. Define

$$K(a,b) = \sum_{\gamma:a\to b} w(\gamma) := \sum_{\gamma:a\to b} \prod_{e\in\gamma} w(e).$$

where the sum is over all paths γ through the graph G starting at a and ending at b. Now consider vertices $a_1,\dots,a_k$ and $b_1,\dots,b_k$ such that there exists at least one tuple of non-intersecting paths $\gamma_1,\dots,\gamma_k$ so that each γ_i is a path from a_i to b_j. Show that

$$\det\big(K(a_i,b_j)\big)_{i,j=1}^k = \sum_{\substack{\gamma_i:a_i\to b_i \\ \gamma_1,\dots,\gamma_k \text{ nonintersecting}}} \prod_{i=1}^k w(\gamma_i).$$

Hint: Recall your proof of the Karlin-Mcgregor formula.

Problem 4 (Dyson Brownian Bridge Absolute Continuity)**.** Let $\mathbf{x},\mathbf{y}\in\mathbb{W}_n$. Let $(\tilde{B}_1,\dots,\tilde{B}_n)$ be n non-intersecting Brownian Bridges on $[0,T]$ conditioned to start at $\mathbf{x}$ and end at $\mathbf{y}$. Let $(B_1,\dots,B_n)$ be a Dyson Brownian Motion started at $\mathbf{x}$ (i.e. it has law $\mathbb{P}^\infty_{\mathbf{x}}$ from the previous question).

Fix $0<s<T$. Let $\mathbb{P}_s$ and $\tilde{\mathbb{P}}_s$ be the law of B and $\tilde{B}$ on $[0,s]$ respectively. Show that

$$\frac{d\tilde{\mathbb{P}}_s}{d\mathbb{P}_s} = \frac{q_{T-s}(B_s,\mathbf{y})}{q_T(\mathbf{x},\mathbf{y})} \le \frac{C_n(T-s)^{-n^2/2}}{q_T(\mathbf{x},\mathbf{y})}$$

where

$$q_t(\mathbf{x},\mathbf{y}) = \frac{\det(p_t(x_i,y_j))_{i,j=1}^n}{\Delta(\mathbf{x})\Delta(\mathbf{y})}, \qquad C_n = \frac{1}{(2\pi)^{n/2}\prod_{j=1}^{n-1} j!}$$

Remark: You may assume that $q_1 \le C_n$ on $\overline{\mathbb{W}}_n\times\overline{\mathbb{W}}_n$

Problem 5 (Brownian Melon and Dyson Brownian Motion)**.** Let $W=(W_1,\dots,W_n)$ be Dyson Brownian motion started from 0, and let $B=(B_1,\dots,B_n)$ be the n-curve Brownian melon on $[0,t]$. Show that

$$(B_1(s),\dots,B_n(s))_{0\le s\le t} \overset{d}{=} \left(\frac{t-s}{\sqrt{t}}W_1\left(\frac{s}{t-s}\right),\dots,\frac{t-s}{\sqrt{t}}W_n\left(\frac{s}{t-s}\right)\right)_{0\le s\le t},$$

where the value at $s=t$ is interpreted by continuity.

Hint: First prove the one-dimensional Brownian motion to Brownian bridge time-change. Then apply it coordinatewise and note that the time-change preserves ordering. To justify the conditioning, first condition on non-intersection on $[\epsilon,t_0]$, then send $\epsilon\downarrow 0$ and $t_0\uparrow\infty$.